Observation - Week 04

Topic: MongoDB and Mongoose

Name: _____ Roll No: _____ Date: _____

Aim

To perform CRUD operations on a MongoDB database using Mongoose models and controllers.

Concepts Covered

- Documents and collections (NoSQL)
- Mongoose schemas, validation and timestamps
- Controllers separating business logic from routes
- CRUD: create, find, findByIdAndUpdate, findByIdAndDelete

Observations

- Schema validation rejects documents that miss required fields before they reach the database.
- unique: true creates an index; duplicate emails raise an error.
- findByIdAndUpdate needs { new: true } to return the updated document.
- MongoDB Compass shows documents stored as BSON with an auto-generated _id.

Viva Questions

1 SQL vs NoSQL?

2 What is an ObjectId?

3 What is a Mongoose schema vs model?

4 What is indexing?

Reference Files

- src/dbConfig.js
- src/models/User.js
- src/controllers/userController.js
- screenshots/mongodb-compass-db.png
- screenshots/crud-operations-output.png